

Technical Whitepaper: Model Fine-Tuning vs RAG Architecture

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Executive Overview:

Fine-tuning adjusts internal neural network weights, whereas RAG provides an external, dynamic non-parametric memory store without model retraining.

For rapidly evolving enterprise documents, RAG is 20x more cost-effective than continuous fine-tuning.

Section 1: Data Freshness and Cost Analysis

Fine-tuning costs approximately \$450 to \$1,800 per training run on specialized clusters.

In contrast, vector ingestion with DocuMind costs less than \$0.002 per document.

Section 2: Hallucination Rates and Verifiability

Fine-tuned models still hallucinate in 18% of domain-specific edge cases because facts are stored implicitly in weight distributions.

RAG solves this by forcing the model to cite exact source excerpts and page numbers.

Contrasting Perspective on Page 2:

While fine-tuning is superior for teaching specific linguistic styles or strict code formats, RAG is the undisputed standard for document discovery, compliance, and factual research.

Section 3: Hybrid Deployment Guidelines

The ideal enterprise architecture combines a fast, compact foundation model (such as gemini-2.5-flash) with dynamic RAG vector indexing.

Operational Conclusions:

1. Use RAG for real-time document search, Q&A, and policy adherence.
2. Avoid static fine-tuning when source PDFs change weekly or monthly.
3. Enable multi-document comparative indexing to detect discrepancies across contracts.